Mohammed Khalid Shaik

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Software engineer with 6 years of experience building scalable systems for data-intensive and AI applications. Shipped security and observability features for an enterprise AI agent harness used in life sciences research. Previously owned large-scale data infrastructure at Takeda and Crinetics. Also supervised their AWS disaster recovery under GxP audit. *(Open to relocation)*

SKILLS

Languages

- Python
- Rust
- JavaScript/TypeScript
- SQL

Frontend

- Tauri
- React

Backend & APIs

- FastAPI, SQLAlchemy
- REST, WebSockets
- Async & concurrent programming
- Microservices
- Distributed systems

Compliance

- GxP
- Change control
- Audit readiness

Data

- PostgreSQL
- Redis
- Pandas
- Pydantic
- Airflow

Cloud & Infra

- AWS
- Terraform
- Kubernetes
- Docker

AI / LLM Engineering

- Sandboxed agent execution
- Pydantic AI orchestration
- LiteLLM routing
- LangCache, LangChainWiki
- RAG with haystack, LlamaIndex

Testing & Quality

- pytest, Locust
- Integration & E2E testing
- AST static analysis
- CI/CD

PROFESSIONAL EXPERIENCE

Software Engineer, Integrated Analytics Solutions — **Remote, May 2023 – Present**

Sentinel, AI agent harness for life sciences research

- **Least-Privilege Agent Sandboxing:** Built least-privilege sandboxing for coding agents with macOS Seatbelt on Sentinel, a life-sciences AI agent platform for drug research. Each agent can reach only the files, tools, MCP servers, and context it's allowed to access.
- **Multi-Agent Patch Pipeline:** Designed a multi-agent system for shipping production fixes and security patches. Workers run isolated in Docker Sandboxes with managed context and observability, and no merge lands until a read-only verifier agent produces test proof. Cut agent errors by 50% and tripled patch releases.
- **Agent Observability:** Extended Codex and OpenCode support to the harness agent monitor, so it tracks their token spend and blocked sessions, too.

Crinetics Pharmaceuticals, clinical-data delivery

- **Regulated Data Delivery:** Rebuilt Crinetics' regulated clinical-data delivery (SFTP to SMB) as an agent-operable service that AI agents and administrators can configure, monitor, diagnose, and recover. It handles thousands of files a day and delivers 100-file bursts in under 3 seconds with exactly-once delivery and immutable GxP audit trails.

Takeda, R&D data infrastructure

- **AWS Disaster Recovery:** Owned AWS disaster recovery for a single-region R&D SaaS platform (10,000+ users, 20 TB), delivering a GxP-validated 2-hour RTO / 4-hour RPO. Authored the standby environment in Terraform and Lambda automation that auto-attached backup EBS volumes to recovery instances at failover.
- **Infrastructure-as-Code Governance:** Served as the approving reviewer on the data infrastructure Terraform repository, where every applied plan mutated AWS infrastructure. Gated 50+ PRs with validated testing, documented releases, and change-control sign-off, enabling junior engineers to ship safely in a GxP environment.

- **FDA Submission Automation:** Built and owned an enterprise document-assembly service (FastAPI) that compiles clinical study RTF outputs into one FDA submission-ready PDF, using a spreadsheet control file to set section order, a clickable table of contents, and bookmarks. Assembly time fell by 97%, from 15+ minutes to 30 seconds per file.
- **Secure Multi-User Architecture:** Engineered the FDA document-assembly service for regulated multi-user production use: isolated per-session storage, file integrity checks, and async processing for up to 100 concurrent sessions and 1 GB workloads. It ran under GxP change control with zero cross-session data leaks, supporting 10,000+ users.

Software Engineer, GameStop **Remote, Nov 2022 – May 2023**

- **Workflow Automation:** Automated promotion approvals with Python, SQL Views, and REST APIs, cutting monthly review effort by 80% (100 hours → 20 hours) and eliminating the manual gaps that had previously exposed the team to compliance penalties.
- **RAG Chatbot:** Built an internal retrieval-augmented generation (RAG) chatbot over company documents, using Haystack for indexing, embedding, and retrieval, and PostgreSQL with pgvector for vector search.

Software Engineer, M&G **Remote, Jan 2021 – Aug 2022**

- **Self-Healing Monitoring:** Developed a service for monitoring and self-healing across 1,500+ servers. Resolved half of recurring tickets automatically, saving 100+ team hours per month and protecting SLA timelines.
- **Identity & Access Management:** Managed identity and access management (IAM) for 100+ users across 15 teams, provisioning and maintaining their access to applications, servers, storage, and other IT infrastructure.

Data Scientist Intern, INSOFE **Hyderabad, June 2020 – Dec 2020**

- **Demand Forecasting:** Built product demand forecasting models in R on Walmart's public retail sales data. Compared the performance of RIMA and ARIMAX time series models against Random Forest, CNN, and LSTM models, and tuned hyperparameters to improve forecast accuracy.

PROJECTS

- **Shell.ai Hackathon (Competition)**: Optimized a 50-turbine offshore wind farm layout to maximize annual energy output. [1]
- **Pigeon (Open Source)**: Unified Claude Code, Codex, and OpenCode session observer. [2]
- **quickxorhash (Open Source)**: Go CLI computing Microsoft QuickXorHash for OneDrive hash parity in production migrations. [3]

EDUCATION

- The University of Texas, *Master's in Data Science* Arlington, TX | Dec 2023
- Jawaharlal Nehru Technological University, *Bachelor of Technology* Hyderabad, India | Aug 2020